

SENSITIVITY OF AN AUDIO-DEEFAKE LEADERBOARD TO THE RESAMPLING UNIT AND TO SOURCE, SPEAKER, AND ATTACK WEIGHTING

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ABSTRACT

Audio-deepfake leaderboards rank detectors from repeated trials that share speakers and spoofing attacks. We analyze eight systems on ASVspoof 2021 DF, giving 28 pairwise contrasts. For systems A and B , $\Delta\text{EER}_{A,B} = 100(\text{EER}_A - \text{EER}_B)$ percentage points; a pair is separated when its 95% simultaneous max- t band excludes zero. A trial-level bootstrap separates 26/28 pairs and 5/6 organizer-baseline pairs, whereas a speaker $\times$ attack product-weight bootstrap separates 18/28 and 0/6. On single-source SpoofCeleb, an analysis archived before dataset access changes the corresponding count from 6/6 to 3/6, showing that between-source mixture changes are not necessary. Twelve fixed positive source/speaker/attack weighting rules reverse 6/12 within-cohort orderings but 0/16 between-cohort orderings. Across the tested perturbations and source deletions, separation between the four SSL detectors and four organizer baselines is stable, whereas their within-cohort ordering is not. These are fixed-score sensitivity bands, not population confidence intervals.

Index Terms— Anti-spoofing, audio deepfake detection, evaluation methodology, clustered bootstrap, ranking uncertainty

1. INTRODUCTION

In 2021 the ASVspoof organizers tested every pair of submitted systems for significance, applied a Holm–Bonferroni correction, and published the deepfake task’s systems in four groups, noting that the top two were not separated [1]. We found no uncertainty analysis in three subsequent benchmark reports: the journal overview [2], ASVspoof 5 [3], and the recent multi-benchmark leaderboard [4] contain none of *significance test*, *confidence interval*, *bootstrap*, *p-value* or *standard error*. The sub-point margins on that modern leaderboard are therefore reported without an uncertainty test; a 2026 benchmark of self-supervised models likewise ranks point EERs without an interval, error bar or test [5]. The 2026 edition adds a calibration study instead; calibration

establishes whether a score means what it says, not whether two systems differ.

That matters more than it would if the analyses the field did run had represented the dependence. ASVspoof 2017 reported 95% EER confidence intervals and read their overlap [6], and 2021 tested every pair [1]; both use [7], and the practice then lapsed. Both variants of the cited test [7] treat trials within a class as independent Bernoulli draws; the dependent form accounts for two systems sharing trials, not for trials sharing speakers and attacks. ASVspoof 2021 DF contains 533,928 evaluation trials but only 93 speakers and 110 attacks, and per-attack EERs for SSL-AASIST span 0.4–10.7%. Across the eight detectors, the one-way random-intercept ICC $\sigma_s^2/(\sigma_s^2 + \sigma_e^2)$ on bona-fide scores ranges from 0.15 to 0.77, where σ_s^2 and σ_e^2 are the speaker-intercept and residual variances. ICC alone does not establish dependence in paired EER differences; that is measured directly below by matched perturbations of the joint scores and checked on two external datasets.

We audit two evaluation choices on fixed released scores: whether trials or speakers and attacks are resampled, and how observed source, speaker and attack groups are weighted. The primary question is whether pairwise separation survives those choices; we propose no new bootstrap and claim no identified population sampling law. On 21DF the six organizer-baseline pairs change from five separated to none, and on single-source SpoofCeleb the corresponding four-detector comparison changes from six to three. Alternative positive weighting rules then localize the stable conclusion: separation between the SSL-detector and organizer-baseline cohorts survives, whereas within-cohort ordering does not.

Scope. The primary set comprises eight score files released by the corresponding model authors or ASVspoof organizers, not the universe of public re-scores. We analyze all eleven Speech DF Arena re-scores [4] separately. Four VCC spoof strata contain only 4/4/4/6 speakers, so no 21DF separation result transfers to new speakers or attacks. The bands and reweighting results describe fixed scores under declared procedures, not population confidence or safety.

2. RELATED WORK

This benchmark’s own reporting. ASVspoof 2021 Fig. 4(c) pairs systems with the Bengio–Mariéthoz test [7] under Holm correction and publishes group structure [1]; Wang and Yamagishi [8] showed that changing only the training seed moves ADD rankings—a variance component our intervals do *not* include.

Clustered resampling. Poh, Martin and Bengio [9] showed that biometric uncertainty is governed by users rather than sample count; Wu et al. carried user-aware resampling into speech [10] and developed a paired correlation-corrected two-system test [11]. Bolle et al. proposed a subsets bootstrap [12], while Fogliato et al. found undercoverage for dyadic 1:1 matching [13]. Our trials instead have one speaker and, for spoof, one attack: two crossed factors [14], a delicate design [15, 16] distinct from nested metrics [17]. Speech bootstrap intervals also include utterance- and block-based forms [18, 19]. Our contribution is the audited fixed-data result: after EER threshold refitting and all-pair multiplicity are aligned, outputs change with the declared perturbation unit. We claim neither cell-level agreement with Fig. 4(c) of the ASVspoof 2021 report nor population confidence.

Evaluation-design audits. Zhuang et al. [20] compute a paired-binomial detectable effect for quantization benchmarks, treating items as independent—the assumption under test here. In audio deepfakes, aggregation weights change conclusions [21], channel presentation can dominate model choice [22], and other audits question representations [23] and corpus overlap [24]. None studies pairwise ranking changes under both dependence-aware resampling and alternative source/speaker/attack weights. Simultaneous rank intervals are established [25]; adjacency is score-selected, so we control multiplicity over all 28 contrasts.

3. METHOD

3.1. EER contrast and separation event

Two deterministic systems on a fixed trial list have no sampling uncertainty until a perturbation law is declared. Scores are oriented so larger means bona fide and sorted in ascending order. At each ordered position, bona-fide mass at or below it is the false-reject rate (FRR) and spoof mass above it is the false-accept rate (FAR); the first position minimizing $|FRR - FAR|$ is selected and EER is their mean, without interpolation. A distinct-score-boundary recomputation changes no displayed value by 0.001 point. For ordered pair (A, B) , $\Delta EER_{A,B} = 100(EER_A - EER_B)$ percentage points (negative favors A); every replicate refits both thresholds. A pair is *separated under procedure h* when its simultaneous band excludes zero; write this indicator as $I_{p,h}$. The identified object is the change in ΔEER and $I_{p,h}$ on the fixed scores, not population uncertainty.

3.2. Trial and speaker $\times$ attack bootstraps

The trial bootstrap independently resamples trials within class. In the speaker $\times$ attack product-weight (PW) bootstrap [16], speakers and spoof attacks are independently resampled with replacement; a spoof trial receives the product of its speaker and attack multiplicities, while a bona-fide trial receives only its speaker multiplicity. Both procedures use the same scores, threshold rule, $B=5000$ shared draws and all-pair multiplicity correction.

3.3. All-pair multiplicity for procedure bands

For procedure h , let $s_{p,h} = \text{sd}_b(\Delta_{p,h}^{(b)})$. We set $q_{.95,h}$ to the empirical 95th percentile of $\max_{p=1}^{28} |\Delta_{p,h}^{(b)} - \bar{\Delta}_{p,h}|/s_{p,h}$ and report $\hat{\Delta}_p \pm q_{.95,h}s_{p,h}$ [26]. This controls all 28 pairwise contrasts within each procedure. Because a population sampling law is not identified for 21DF, we call these procedure-sensitivity bands, not confidence intervals.

Reconstructing the cited test. Fig. 4(c) of Yamagishi et al. [1] does not disclose which Bengio–Mariéthoz variant produced its matrix, so we make no cell-level attribution. With each system’s non-interpolated EER threshold selected on the evaluation set, our independent-trial and shared-trial reconstructions separate the same five of six baseline pairs. The published 34-system matrix has $\binom{34}{2} = 561$ tests; even its Bonferroni cutoff is exceeded by the weakest of our five reconstructed results ($p = 9.7 \times 10^{-6}$), while the sixth has $p = .217$. The cited derivation assumes thresholds fixed outside the test set, so these are reconstructions of its adaptation to EER rather than exact finite-sample tests.

3.4. Alternative fixed weighting rules

Within each class, every named hierarchy level receives equal conditional mass and trials divide their terminal-group mass equally. Bona-fide rules are empirical trial weights, equal mass over the three sources, and equal source $\rightarrow$ speaker mass. Spoof rules are empirical trial weights, equal mass over five spoof strata, equal mass over observed speaker $\times$ attack cells within stratum, and equal mass successively over stratum, attack family, attack and speaker. Their 3×4 crossing gives 12 positive, class-normalized rules fixed without inspecting scores. For endpoint w^1 , the path from empirical weights is $w(\lambda) = (1 - \lambda)w^0 + \lambda w^1$, $0 \leq \lambda \leq 1$. We measure its distance by maximum classwise total variation, $\max_c \frac{1}{2} \sum_i |w_{c,i} - w_{c,i}^0|$. Complete formulas, group maps, simulations and all 28 numerical bands are in the versioned supplement.

4. EXPERIMENTS

Setup. The primary set contains four author-released self-supervised systems (XLSR-Mamba [27], XLS-R+SLS

[28], XLSR-Conformer [29], SSL-AASIST [30]) and four organizer-released baselines (RawNet2, LFCC-LCNN, LFCC-GMM, CQCC-GMM). We use all 533,928 evaluation-phase trials for which every score is available. Table 1 reports the conventional point results. Pairwise bands use seed 20260817.

Primary 21DF result (Fig. 1). Both reconstructions of Fig. 4(c) separate the same five of six organizer-baseline pairs; we claim no cell-level agreement with the published matrix. The matched all-28-pair bootstrap refits the same weighted EER in every replicate. Trial resampling separates 26/28 pairs and 5/6 organizer-baseline pairs; the PW bootstrap separates 18/28 and 0/6. A constrained PW bootstrap that exactly restores each of the three bona-fide source masses and five spoof-stratum masses also separates 18/28 and 0/6. This constrained arm was designed after an earlier arm failed; it is a robustness check, not prospective evidence. Thus neither threshold handling nor movement of these eight class-specific masses is necessary for the contrast. For RawNet2–CQCC-GMM, $\Delta\text{EER} = -3.180$ points and the PW band is $[-9.284, 2.925]$.

Single-source SpoofCeleb check. Repository commit `c2efa63`, dated 29 August 2026 while official access was pending, fixed the four detectors, scoring transforms, bootstrap laws, seed and six-pair endpoint before the first successful authenticated download. On all 91,130 evaluation trials [31], trial resampling separates 6/6 pairs and the PW bootstrap separates 3/6. AASIST, SLS, SSL-AASIST and XLSR-Mamba have EERs 57.93%, 24.51%, 26.72% and 27.58%; this is an off-domain sensitivity check, not a comparison of competitive SpoofCeleb systems. Every base item has one bona-fide and nine spoof renditions under one source label, so between-source mass cannot move; speaker and spoof-attack multiplicities still vary under PW draws. After observing this result, separate speaker-only and attack-only bootstraps separated 5/6 and 3/6; attack-only matched the PW indicator vector. The archived plan did not bind the scoring executable, so a later clean-source re-scoring is reproducibility evidence only: three score files were byte-identical, the fourth differed by at most 1.44×10^{-6} , and every EER and separation indicator was unchanged.

What 21DF supports. The PW bootstrap separates all 16 SSL-versus-baseline pairs, 2/6 within-SSL pairs and 0/6 within-baseline pairs. The 16 cross-cohort gaps are 19.5–23.7 points and remain separated after each leave-one-source-out refit. The two PW-separated within-SSL gaps (0.968 and 0.935 points) cease to separate after a source deletion; adjacent within-SSL gaps are only 0.032, 0.357 and 0.578 points. PW bands are $4.13\text{--}11.36\times$ wider than matched trial bands over all pairs (median $8.14\times$).

Alternative weights. The 12 fixed rules reverse 6/12 within-cohort orderings and 0/16 cross-cohort orderings; one of the 11 nonempirical linear paths reverses before maximum classwise total variation .10. After observing these out-

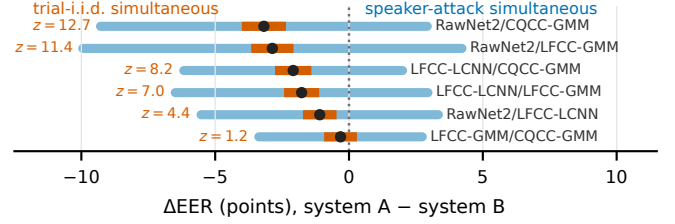


Fig. 1. The six organizer-baseline pairs, with matched all-28-pair max- t bands. Narrow: simultaneous trial-bootstrap bands; wide: simultaneous PW-bootstrap bands. The independent-trial reconstruction’s z at left separates the same five pairs as the narrow bands. Dots are point estimates; no wide band excludes zero.

Table 1. Point EER (%) on the 21DF evaluation set. “Release” identifies the score-file source. Rank is point order only, not an uncertainty claim.

System	Release	EER (%) ↓	Rank
XLSR-Mamba	author	1.884	1
XLS-R+SLS	author	1.916	2
XLSR-Conformer	author	2.273	3
SSL-AASIST	author	2.852	4
RawNet2	organizer	22.383	5
LFCC-LCNN	organizer	23.477	6
LFCC-GMM	organizer	25.247	7
CQCC-GMM	organizer	25.563	8

comes, a constrained search found a positive weighting at distance .010218 that changes XLSR-Mamba minus XLS-R+SLS from -0.032130 to $+0.000699$ EER points. This is a found upper bound, not a minimum, and is too small to imply practical superiority. The search found no cross-cohort reversal, which is not a proof that none exists.

Independent re-scores. On the eleven complete Speech DF Arena score files, trial and PW bootstraps separate 52/55 and 38/55 pairs; the median pointwise-band width ratio is 8.45. These are descriptive results for that separate dataset, not population confidence intervals.

Source deletions. The bona-fide sources ASVspoof, VCC2018 and VCC2020 contain 67, 12 and 14 speakers. Leave-one-source-out refits preserve separation for all 16 cross-cohort pairs, remove both separated within-SSL pairs and separate three within-baseline pairs in at least one deletion. Thus cross-cohort separation survives these source changes, while every within-cohort result is conditional on them.

ASVspoof 5 check. For SSL-AASIST, AASIST, XLS-R+SLS and XLSR-Mamba on all 680,774 Track 1 trials [3], role-stratified speaker and attack resampling removes at least one trial-bootstrap separation and gives a median simultaneous-band width ratio of $27.27\times$, exceeding the declared $2\times$ threshold. This check is intentionally EER-

only because it transfers the paper’s Δ EER comparison; ASVspooF 5’s primary Track 1 metric is minDCF. It is not a system-performance replication or population claim.

Coverage boundary. Under imposed bivariate random-effects models on the 21DF trial-to-group structure, the minimum low-EER coverages of the raw exact-cell inclusion–exclusion jackknife, its componentwise variance-floor version and the PW-percentile band are .845, .845 and .855; all three are below .90 in the same 11/24 low-EER cells. The complete 48-cell parameters and method formulas are in the supplement. This simulation evaluates procedures under those imposed models; it does not establish that either model describes 21DF.

5. DISCUSSION

Reporting and limits. The complete 28-row band table, source/stratum map, weighting formulas, simulation design and artifact identifiers are in SUPPLEMENT.md at repository tag ml-field-native-r2-20260830. The legacy SSL-AASIST/AASIST files lack their originating run logs. Benchmark reports should publish trial-to-speaker/attack membership and source weights, not only trial counts. A band containing zero means the procedure does not separate that pair; it does not establish equality. We make no population, rank-confidence or formal Arena-inference claim.

6. CONCLUSION

On fixed released scores, matched 21DF bootstraps change the separated organizer-baseline count from 5/6 to 0/6, and the pre-access-archived single-source SpoofCeleb check changes it from 6/6 to 3/6. Because SpoofCeleb has one source, between-source movement cannot explain the latter contrast; an attack-only bootstrap computed after the result matches its 3/6 indicator vector. The 12 fixed weighting rules reverse 6/12 within-cohort orderings but 0/16 cross-cohort orderings. Across these rules, source deletions and searched directions, separation between the four SSL detectors and four organizer baselines is stable; within-cohort ranking is not. Population confidence requires independently sampled units and a defensible acquisition law.

7. REFERENCES

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